

Dim Light Melatonin Onset (DLMO) and Functional Connectivity in Later-Life Adults

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Introduction

- Around 7.2 million U.S adults aged ≥ 65 live with Alzheimer disease & related dementias (ADRD), which increases as the population ages.
- ADRD disrupts resting-state functional connectivity, the at-rest BOLD activity between brain regions, measured with resting-state functional MRI (rs-fMRI).
- The Default Mode Network (DMN) supports memory & self-referential processing; The Salience Network (SN) switches attention between internal and external directed thoughts.
- Both networks are disrupted in ADRD, while circadian disruptions are linked with accelerated brain ageing and neurodegenerative risk.

Dim Light Melatonin Onset

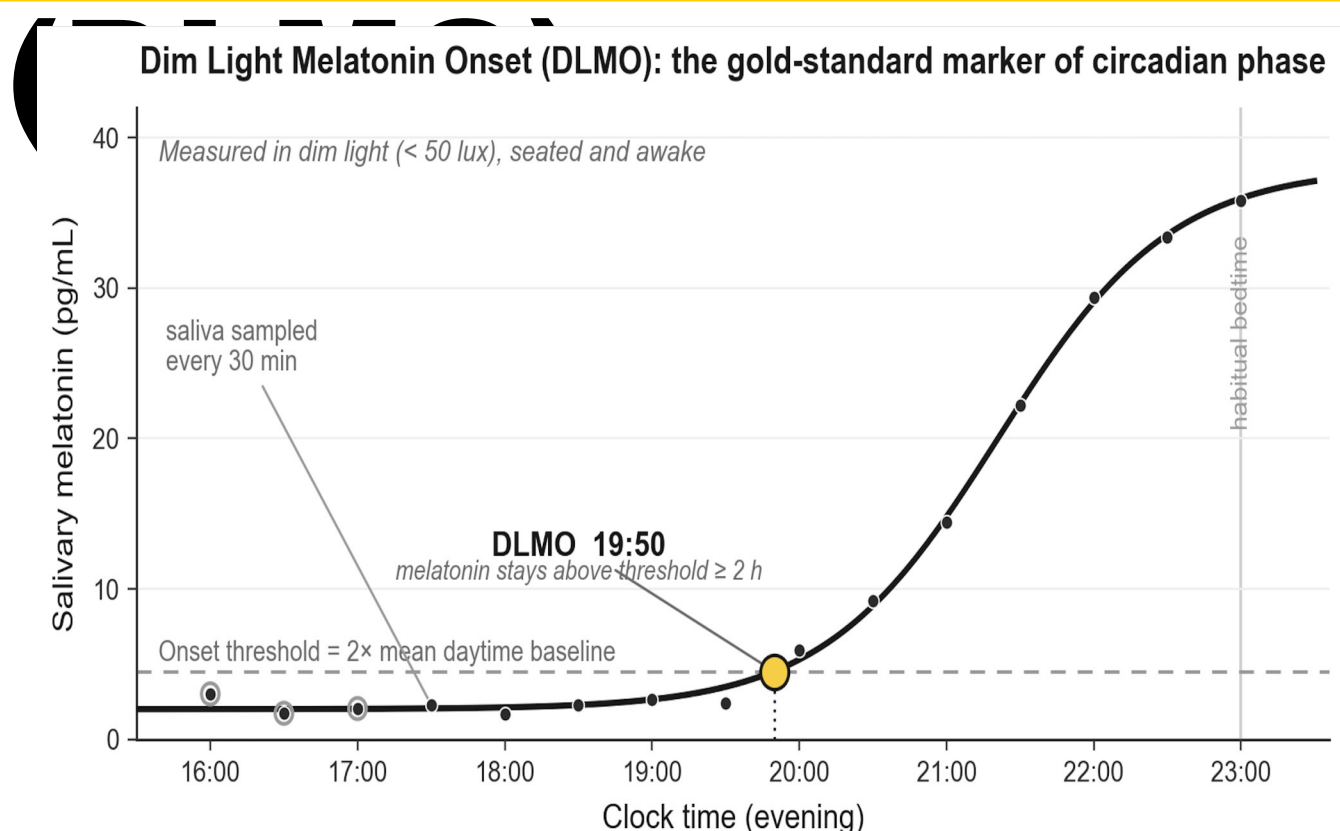


Figure 1: DLMO = clock time measured in dim light (<50 lux), when saliva melatonin crosses 2x daytime baseline threshold and stays elevated for ≥ 2 hr. Early DLMO median 18:45; Late DLMO median 20:30.

Research Question

To determine the relationship between circadian timing and resting-state functional connectivity in hippocampus-DMN, hippocampus-SN, as well as within DMN and SN.

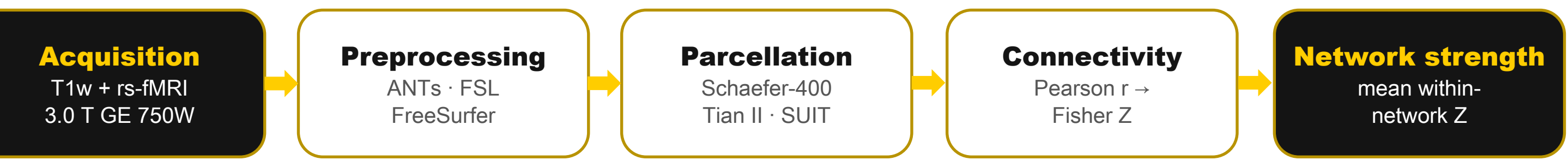
Hypothesis

We hypothesized that in cognitively healthy later-life adults, later circadian rhythm is associated with stronger resting-state functional connectivity in the DMN and SN, the networks often affected in aging and ADRD. Since aging progresses circadian rhythms (earlier DLMO) and weakens connectivity, a later DLMO is expected to have stronger connectivity.

- Later DLMO is associated with greater connectivity within the DMN (A, B, C) and between the hippocampus and DMN.
- Later DLMO is associated with greater connectivity within the SN (A, B) and between the hippocampus and SN.

Methods

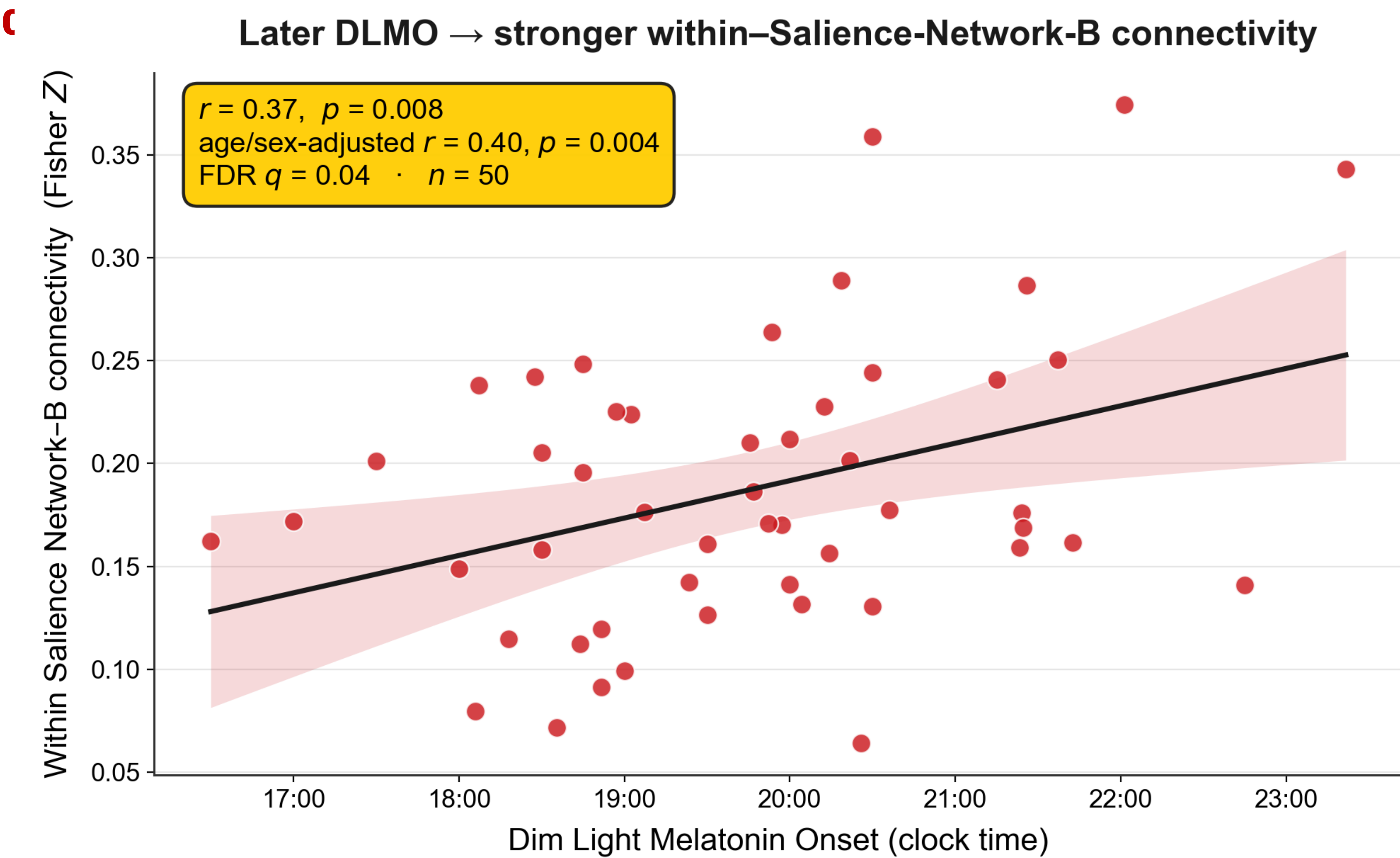
- 52 later-life adults (mean age = 68, SD= 6.50, 52% female) underwent Dim Light Melatonin Onset (DLMO), MRI, and interviews.
- Circadian timing was measured by home DLMO kit. DLMO is recognized as the most accurate marker for measuring circadian timing. DLMO intermittently measures melatonin concentrations in saliva while in dim light (<50 lux).
- T1-weighted MRI and resting-state functional MRI were acquired using 3.0T scanner (GE Discovery). Images were acquired by ANTs, FSL, FreeSurfer, AFNI . The functional connectivity was assessed by combining five different atlases to provide regions of interest including Schaefer 400 atlas, SUI, MAGeT-Brain, Harvard-Oxford cortical and subcortical structural atlas, and CIT168 (**Figure 1**). Then, the Pearson’s correlation values were normalized using Fisher’s Z transformation.
- Descriptive statistics were used to summarize the sample characteristics and multiple regression was used to examine the relationship between DLMO and functional connectivity (i.e., default mode network and salience network).



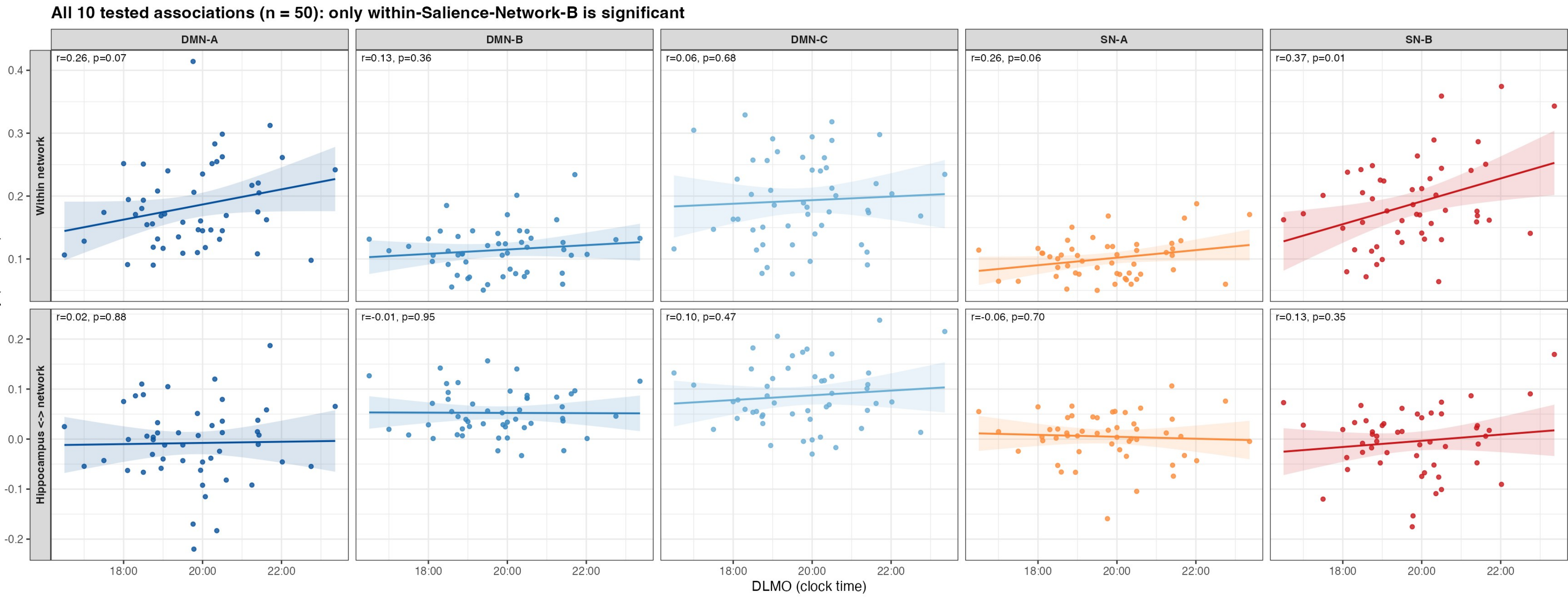
Results

Later DLMO → stronger within-Salience-Network-B connectivity

($r = 0.37$, $p = 0.008$; age/sex-adjusted $r = 0.40$, $p = 0.004$; FDR $q = 0.04$)



Specificity → all 10 tested associations: across both connectivity measures (within-network strength and hippocampus ↔ network) for every subnetwork, only within-SN-B reached significance (all others $p \geq 0.06$).



Why SN-B?

- The dorsal SN (dorsal ACC + anterior insula) manages attention switching between internal (DMN) and external stimuli.
- A delayed circadian rhythm may alter salience attention processing; a plausible early, network-specific pre-clinical signal in healthy aging.

In cognitively healthy later-life adults, a **later circadian rhythm correlates with stronger dorsal-Salience-Network connectivity**. While DMN and hippocampal connectivity are unchanged.
A potential pre-symptomatic, network-specific, marker of circadian rhythms influence on the aging brain.

Table 1. Sample Characteristics				
	Early Melatonin Onset (n = 25)	Late Melatonin Onset (n = 25)	Total (n = 50)	
	N (%) or median (range)			P-value
Female	17 (68%)	9 (36%)	26 (52%)	0.024
Male	8 (32%)	16 (64%)	24 (48%)	
Age	68 (61, 84)	67 (58, 86)	67 (58, 86)	0.62
Education (yrs)	16 (14, 29)	16 (8, 25)	16 (8, 29)	0.032

Early/Late = median split of DLMO (≈19:50).

Conclusions

- Later circadian timing was significantly associated with **higher within-Salience-Network B** connectivity in cognitively healthy later-life adults.
- The observation was **network-specific**: no significant association with within-DMN, within-SN-A, or hippocampus↔network connectivity.
- The SN-B supports attention switching; delayed circadian rhythms may alter salience processing throughout aging.
- Circadian timing may influence functional connectivity, potentially contributing to neuronal changes underlying neurodegenerative diseases.
- A longitudinal design with a larger sample size is needed to determine if circadian rhythms can be used to predict cognitive courses.

Acknowledgments

The first author (Baggot) was supported by the University of Iowa Undergraduate Research Fellowship and University of Iowa Csomay Center (PI: Zhang). This work was supported by University of Iowa Csomay Center (PI: Moon). The University of Iowa Institute for Clinical and Translational Science (NIH/ACTS KL2TR002536, PI: Bassuk, Scholar: Moon), and Alzheimer’s Association (AARG-19-618403, PI: Moon). This work was conducted on an MRI instrument funded by 1S10OD025025-01. Contents are solely the responsibility of the authors and do not necessarily represent the official views of the University of Iowa, NIH/ACTS, or Alzheimer’s Association.

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